Summary in Graph

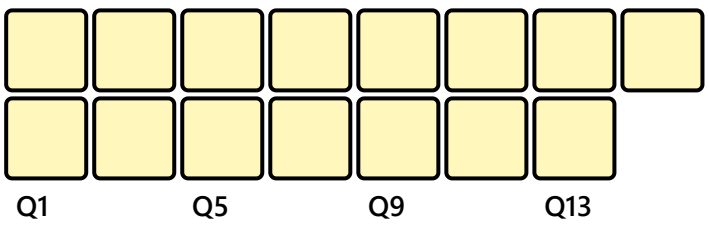
Exam Summary (GO Classes GATE CS | Digital Logic | Number System | Weekly Quiz 1)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	15 (5 + 10)
Total Marks:	25 (5 + 20)
Exam Duration:	45 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (37) out of 373 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Technical

Q #1 | Numerical Type | Award: 1 | Penalty: 0 | Digital Logic

Let X be the number of distinct 16-bit integers in $2's$ complement representation. Let Y be the number of distinct 16-bit integers in sign magnitude representation Then $X - Y$ is_____.

Your Answer: | Correct Answer: 1 | Not Attempted | Time taken: 00min 01sec | Discuss

Q #2 | Multiple Select Type | Award: 1 | Penalty: 0 | Digital Logic

Zero has two representations in

A. Sign-magnitude

B. $2's$ complement

C. $1's$ complement

D. None of the above

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Digital Logic

Given that numbers A and B are two 8-bit 2's complement numbers with $\mathrm{A} = 11111111,\ \mathrm{B} = 11111111.$ Then sum $A + B$ is

A. 00000010

B. 11111100

C. 11111110

D. 00000000

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Digital Logic

The $2's$ complement representation of the number $(-100)_{10}$ in an 8 bit computer is

A. 10011011

B. 01100100

C. 11100100

D. 10011100

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Digital Logic

The smallest integer that can be represented by an 8-bit number in $2's$ complement form is

A. -256

B. -128

C. -127

D. 0

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Select Type | Award: 2 | Penalty: 0 | Digital Logic

The number -6 can be represented as 1010 in 4-bit 2's complement representation. Which of the following is/are CORRECT 2 's complement representation(s) of -6?

A. $1000\ 1010$ in 8 -bits

B. $1111\ 1010$ in 8-bits

C. $1000\ 0000\ 0000\ 1010$ in 16-bits

D. $1111\ 1111\ 1111\ 1010$ in 16-bits

Your Answer: | Correct Answer: B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Digital Logic

Which of the following is the representation of the decimal number (-147) in $2's$ compliment notation on a 12-bit machine?

A. 111101101100

B. 110001001101

C. 111101101101

D. 000001101101

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Digital Logic

The maximum and minimum value represented in signed 16-bit $2's$ compliment representation are

A. -32768 and 32767

B. 0 and 32767

C. 0 and 65535

D. -16384 and 16383

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Digital Logic

In an 8-bit representation of computer system, the decimal number 47 has to be subtracted from 38 and the result in binary $2's$ complement is

A. 11110111

B. 10001001

C. 11111001

D. 11110001

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Numerical Type | Award: 2 | Penalty: 0 | Digital Logic

In an 8 bit representation of computer system the decimal number 26 has to be subtracted from 22 and the result in binary 2's complement is — — — — — — —

Your Answer: | Correct Answer: 11111100 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Digital Logic

Subtract $(1010)_2$ from $(1101)_2$ using first complement

A. $(1100)_2$

B. $(0101)_2$

C. $(1001)_2$

D. $(0011)_2$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Digital Logic

The range of n-bit signed magnitude representation is

A. 0 to $2^n - 1$

B. $-\left(2^{n-1} - 1\right)$ to $\left(2^{n-1} - 1\right)$

C. $-\left(2^n - 1\right)$ to $\left(2^n - 1\right)$

D. 0 to $2^{n-1} - 1$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Numerical Type | Award: 2 | Penalty: 0 | Digital Logic

Determine the base of the numbers for the following operation to be correct: $14/2 = 5$

Your Answer: | Correct Answer: 6 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Numerical Type | Award: 2 | Penalty: 0 | Digital Logic

Determine the base of the numbers for the following operation to be correct: $24 + 17 = 40$

Your Answer: | Correct Answer: 11 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Numerical Type | Award: 2 | Penalty: 0 | Digital Logic

The solutions to the quadratic equation $x^2 - 11\text{x} + 22 = 0$ are $x = 3$ and $x = 6$. What is the base of the numbers?

Your Answer: | Correct Answer: 8 | Not Attempted | Time taken: 00min 00sec | Discuss

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